

CENG381 Stochastic Processes

Chapman-Kolmogorov & Diagonalization — Practice 1 (Answer Key)

Q1 Solution

a) P^2 by direct multiplication:

$$\begin{aligned}P^2(S, S) &= P(S, S) * P(S, S) + P(S, R) * P(R, S) \\&= 0.7 * 0.7 + 0.3 * 0.5 = 0.49 + 0.15 = 0.64\end{aligned}$$

$$\begin{aligned}P^2(S, R) &= P(S, S) * P(S, R) + P(S, R) * P(R, R) \\&= 0.7 * 0.3 + 0.3 * 0.5 = 0.21 + 0.15 = 0.36\end{aligned}$$

$$\begin{aligned}P^2(R, S) &= P(R, S) * P(S, S) + P(R, R) * P(R, S) \\&= 0.5 * 0.7 + 0.5 * 0.5 = 0.35 + 0.25 = 0.60\end{aligned}$$

$$\begin{aligned}P^2(R, R) &= P(R, S) * P(S, R) + P(R, R) * P(R, R) \\&= 0.5 * 0.3 + 0.5 * 0.5 = 0.15 + 0.25 = 0.40\end{aligned}$$

	S	R
S	[0.64 0.36]	
R	[0.60 0.40]	

b) Chapman-Kolmogorov and $P^4 = P^2 \cdot P^2$:

C-K: $P^{(n+m)}(i,j) = \sum_k P^n(i,k) * P^m(k,j)$ (matrix product $P^n * P^m$).

$$\begin{aligned}P^4(S, S) &= P^2(S, S) * P^2(S, S) + P^2(S, R) * P^2(R, S) \\&= 0.64 * 0.64 + 0.36 * 0.60 \\&= 0.4096 + 0.2160 = 0.6256\end{aligned}$$

$$P^4(S, R) = 0.64 * 0.36 + 0.36 * 0.40 = 0.2304 + 0.1440 = 0.3744$$

$$P^4(R, S) = 0.60 * 0.64 + 0.40 * 0.60 = 0.3840 + 0.2400 = 0.6240$$

$$P^4(R, R) = 0.60 * 0.36 + 0.40 * 0.40 = 0.2160 + 0.1600 = 0.3760$$

c) Verification of $P^4(S,S)$:

Direct: multiply row S of P^2 by column S of P^2 :

$$[0.64 \quad 0.36] * [0.64, \quad 0.60]^T = 0.64 * 0.64 + 0.36 * 0.60 = 0.6256$$

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Q2 Solution

a) Eigenvalues:

$$\begin{aligned}\det(P - \lambda I) &= 0 \\ \begin{vmatrix} 0.7-\lambda & 0.3 \\ 0.5 & 0.5-\lambda \end{vmatrix} &= 0 \\ (0.7-\lambda)(0.5-\lambda) - 0.3 \cdot 0.5 &= 0 \\ 0.35 - 1.2\lambda + \lambda^2 - 0.15 &= 0 \\ \lambda^2 - 1.2\lambda + 0.20 &= 0 \\ (\lambda - 1)(\lambda - 0.2) &= 0 \\ \lambda_1 &= 1, \quad \lambda_2 = 0.2 = 1-p-q \quad \checkmark\end{aligned}$$

b) Verification of the formula for $n=1$:

$$\begin{aligned}P^n(S, S) &= q/(p+q) + (1-p-q)^n * p/(p+q) \\ &= 0.5/0.8 + (0.2)^n * 0.3/0.8 \\ &= 5/8 + (3/8) * (0.2)^n \\ n=1: \quad 5/8 + (3/8) * 0.2 &= 5/8 + 3/40 = 25/40 + 3/40 = 28/40 = 0.70 \\ &= P(S, S) \quad \checkmark\end{aligned}$$

c) $P^4(S, S)$ from formula vs Q1:

$$\begin{aligned}P^4(S, S) &= 5/8 + (3/8) * (0.2)^4 \\ &= 5/8 + (3/8) * 0.0016 \\ &= 0.6250 + 0.0006 = 0.6256 \quad \checkmark \quad (\text{matches Q1})\end{aligned}$$

Q3 Solution

a) Stationary distribution:

$$\begin{aligned}\lim_{n \rightarrow \infty} P^n(S, S) &= 5/8 + 0 = 5/8 \\ \pi_S &= 5/8 = 0.625, \quad \pi_R = 1 - 5/8 = 3/8 = 0.375\end{aligned}$$

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Verify $\pi \cdot P = \pi$:

$$\begin{aligned}(\pi * P)_S &= (5/8) * 0.7 + (3/8) * 0.5 = 35/80 + 15/80 = 50/80 = 5/8 \\ \checkmark \\ (\pi * P)_R &= (5/8) * 0.3 + (3/8) * 0.5 = 15/80 + 15/80 = 30/80 = 3/8 \\ \checkmark\end{aligned}$$

b) Smallest n^* such that $|P^n(S,S) - \pi_S| < 0.01$:

$$\begin{aligned}|P^n(S,S) - 5/8| &= (3/8) * (0.2)^n < 0.01 \\ (0.2)^n &< 0.01 * 8/3 = 0.02667 \\ n * \ln(0.2) &< \ln(0.02667) \\ n &> \ln(0.02667) / \ln(0.2) = (-3.625) / (-1.609) = 2.25 \\ \text{Smallest integer: } n^* &= 3 \\ \\ \text{Check } n=3: (3/8) * (0.2)^3 &= (3/8) * 0.008 = 0.003 < 0.01 \quad \checkmark \\ \text{Check } n=2: (3/8) * (0.2)^2 &= (3/8) * 0.04 = 0.015 > 0.01 \quad \times\end{aligned}$$

c) Effect of λ_2 on mixing speed:

$$\begin{aligned}\text{If } \lambda_2 &= 0.9: \\ |P^n(S,S) - \pi_S| &= C * (0.9)^n < 0.01 \\ (0.9)^n &< 0.01/C \\ n &> \log(0.01/C) / \log(0.9) \Rightarrow n^* \sim 44 \text{ steps (much larger)}\end{aligned}$$

When λ_2 is close to 1, the term λ_2^n decays slowly — many steps are needed before the chain 'forgets' its starting state. Geometrically, the chain almost never mixes because the spectral gap $(1 - |\lambda_2|)$ is tiny. A spectral gap of $1 - 0.2 = 0.8$ gives very fast mixing; a gap of $1 - 0.9 = 0.1$ is 8× slower.